

#18

Decision Tree Learning / Gini Impurity / Information gain

Date:
 Page No:

- Utility function
- Decision Tree Learning
- family of Decision tree algorithm —
 - ID3
 - C4.5
 - C5.0

— CART [To build a tree we use DT Learning Algorithm called CART]

There is whole family of algorithm used to build tree from data. It gives procedure to ask which question & when

CART — Classification and Regression Tree

How it works —

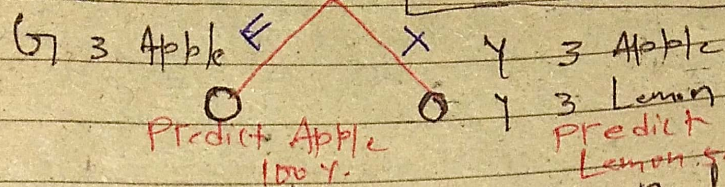
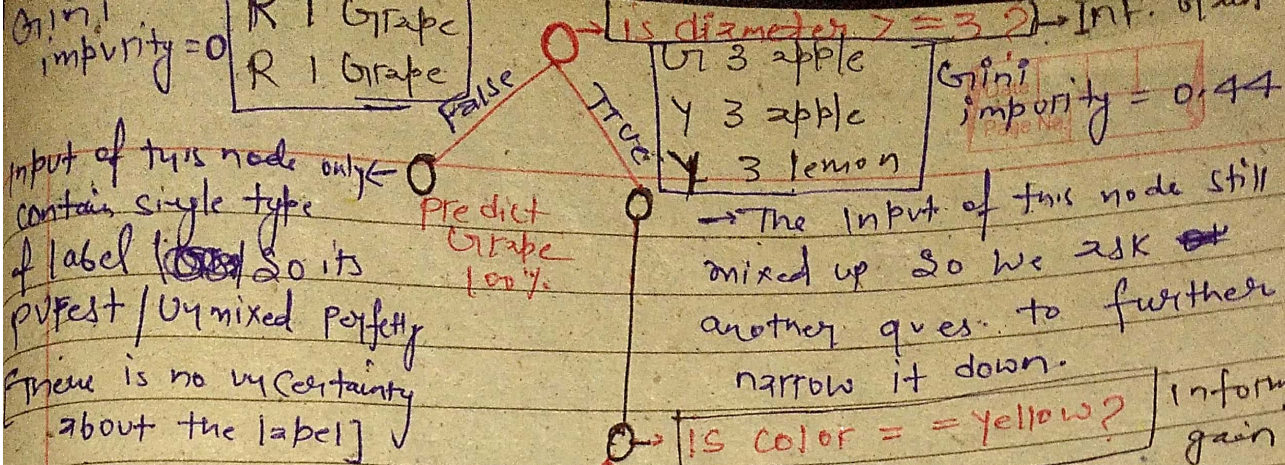
- ① To begin we will add root node for the tree & root will receive entire training set

Color	Diameter	Label
Green	3	Apple
Yellow	3	Apple
Red	1	Grapes
Red	1	Grapes
Yellow	3	Lemon

- ② each node ask T & F ques. about one of the features & in response of this ques. we split, or partition data into two subsets.

- ③ The subset becomes the input ~~for~~ ^{to two} the child node that we add to the tree

- ④ The goal is to produce purest possible distribution of the labels at ~~the~~ each node



✓ The trick to build effective tree we need to know under which question to ask and when? & to do that we need to quantify how much a question helps to unmix the labels.

vi) We can quantify the amount of uncertainty at a single node using a metric called Gini impurity.

vii) We can quantify how much a question reduces that uncertainty using a concept called information gain.

We will use this both concept to ask best ques. at each point

viii) given these concept we will build recursively build the tree on each of the new nodes

ix) We will continue dividing the data until there no further question to ask, at which point we will add leaf.

To implement this we 1st decide what type of question can you ask from data & then about the data.

2nd → How to decide where question to ask & when

quantifying uncertainty

Criterion: Impurity quantify how much uncertainty in the node

Information gain quantify that how much question reduces that

Impurity

Chance of being incorrect if you randomly
Assign a label to an example in the
Same Set

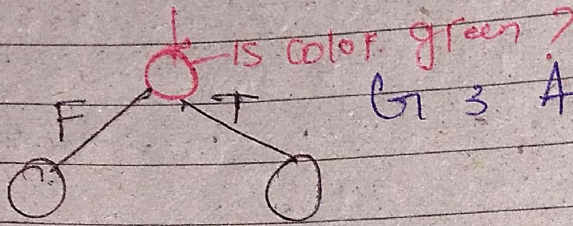
Information gain — Find the
 It's just a number that tells us how much a ques.
 helps to unmix the labels at a node.

Color	Diam	label
G	3	A
Y	3	A
R	1	G
R	1	G
Y	3	L

impurity = 0.64

impurity = 0.62

Y	3	A
R	1	G
R	1	G
Y	3	L



impurity = 0

Calculating the uncertainty of the
 child nodes that result.

Taking weighted avg. of their uncertainty because
 we care more about a large set with low
uncertainty & small set with high.

$$\text{Avg. Imp} = \frac{4}{5} \times 0.62 + \frac{1}{5} \times 0 \approx 0.5$$

starting value

$$\text{Information gain} = 0.64 - 0.5 \rightarrow \text{Avg. uncert.} = 0.14$$

Putting it all together

def build_tree(rows):

info_question = find_best_split(rows)

if info == 0: return leaf(rows)

true_rows, false_rows =

partition(rows, question)

true_branch = build_tree(true_rows)

False_branch = build_tree(False_rows)

Date			
Page No.			

return Decision_Node(question, true_branch, False_branch)

#9